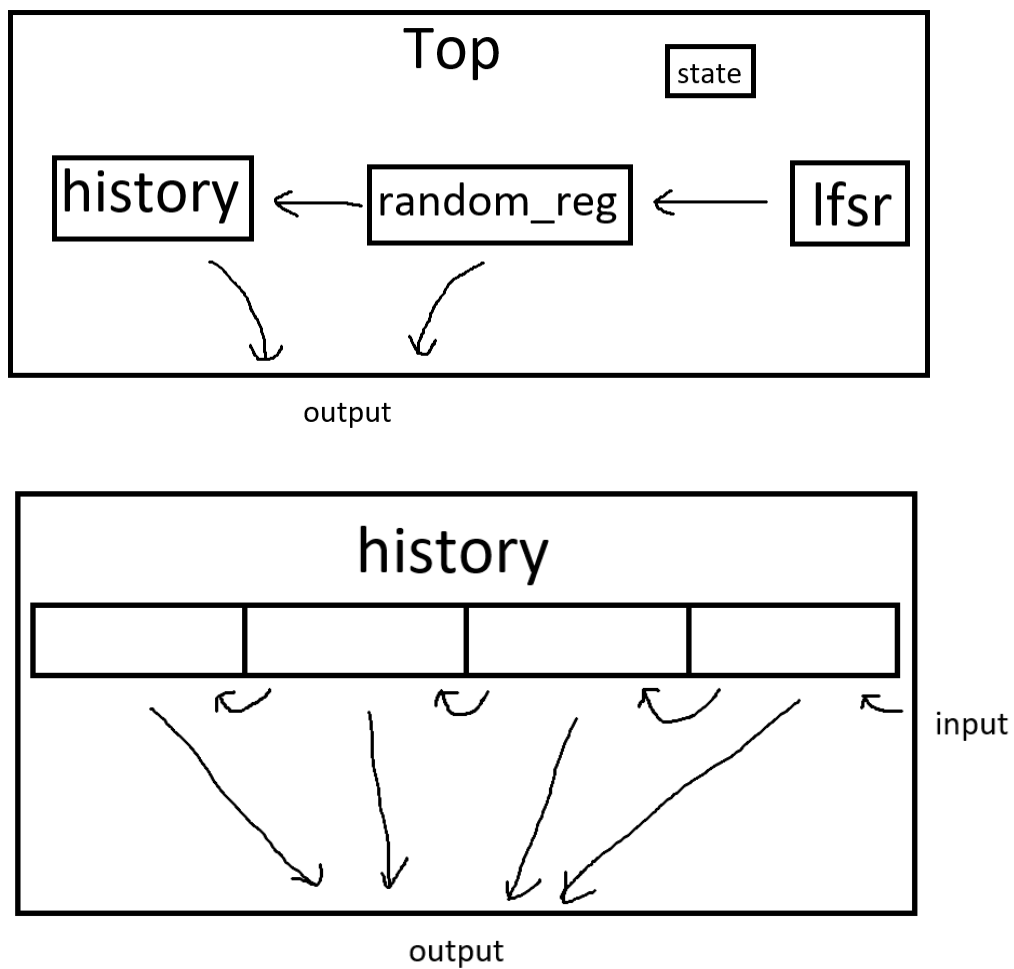


Lab 1 Report

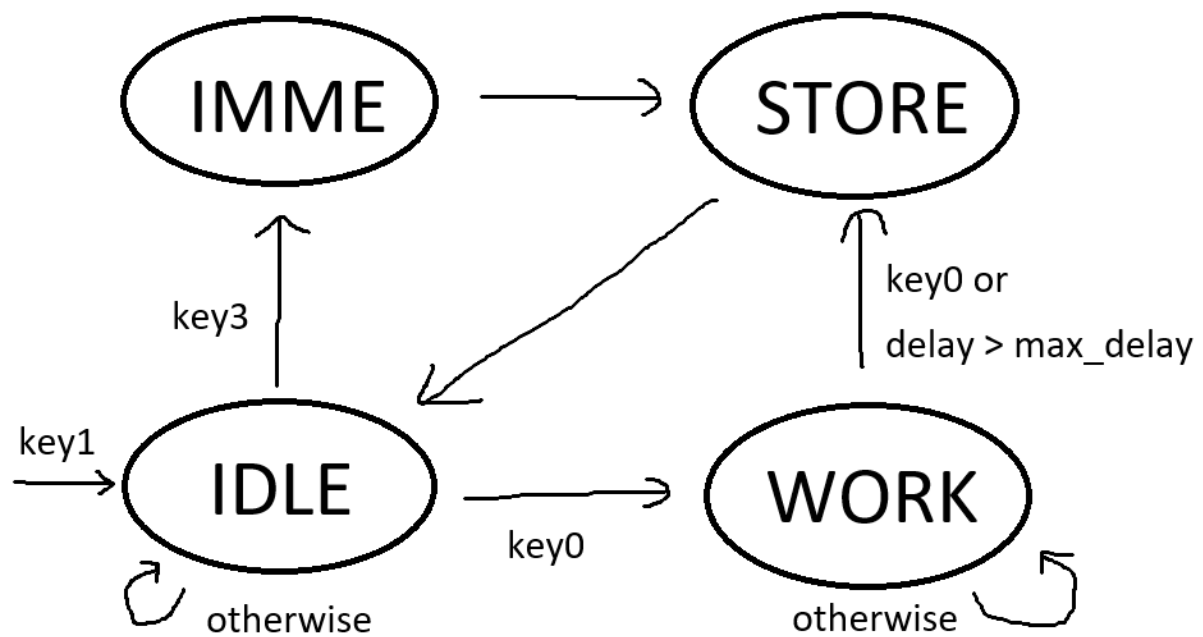
File Structure

```
team10_lab1/  
| - team10_lab1_report.pdf  
| - src/  
| - DE2_115/  
| - DE2_115.qsf  
| - DE2_115.sdc  
| - DE2_115.sv  
| - Debounce.sv  
| - SevenHexDecoder.sv  
| - Top.sv
```

System Architecture



Hardware Scheduling



Fitter Summary

Fitter Summary	
Fitter Status	Successful - Thu Mar 12 20:02:29 2026
Quartus II 64-Bit Version	15.0.0 Build 145 04/22/2015 SJ Full Version
Revision Name	DE2_115
Top-level Entity Name	DE2_115
Family	Cyclone IV E
Device	EP4CE115F29C7
Timing Models	Final
Total logic elements	190 / 114,480 (< 1 %)
Total combinational functions	164 / 114,480 (< 1 %)
Dedicated logic registers	137 / 114,480 (< 1 %)
Total registers	137
Total pins	518 / 529 (98 %)
Total virtual pins	0
Total memory bits	0 / 3,981,312 (0 %)
Embedded Multiplier 9-bit elements	0 / 532 (0 %)
Total PLLs	0 / 4 (0 %)

Timing Analyzer

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Compilation Report - DE2_115

Unconstrained Paths

	Property	Setup	Hold
1	Illegal Clocks	0	0
2	Unconstrained Clocks	0	0
3	Unconstrained Input Ports	4	4
4	Unconstrained Input Port Paths	140	140
5	Unconstrained Output Ports	11	11
6	Unconstrained Output Port Paths	40	40

遇到的問題與解決辦法, 心得與建議

問題: 不知道如何使用 key2 和 key3

解決方法: 改 ./DE2_115/DE2_115.sv

心得: 我發現還有一堆 SW (switch 沒用到)

建議: 希望簡報可以在上課時間就放上 NTU COOL

Bonus

跳動中途立即停止 (再按一次 key0)

記憶前次亂數結果 (含本次共四次) (按 key2 檢視)

瞬間產生一個亂數 (不須等待) (按 key3)

Implementation

32-bit Linear feedback shift register

```
wire new_bit = lfsr_reg[31] ^ lfsr_reg[21] ^ lfsr_reg[1] ^ lfsr_reg[0];  
wire[31:0] lfsr = {lfsr_reg[30:0], new_bit};
```

Seed 不能是 0